SIMATS ENGINEERING
ASSESSMENT TOOL 1: Concept Mapping

COURSE NAME: Cloud Computing and Big Data Analytics **COURSE CODE** :CSA1522

Name	**MAGESH.S**
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COURSE OUTCOME COVERED
CO1: *Apply cloud computing technologies including hardware-software evolution, virtualization, web services, and service models to develop cloud-based solutions.* (BL2)

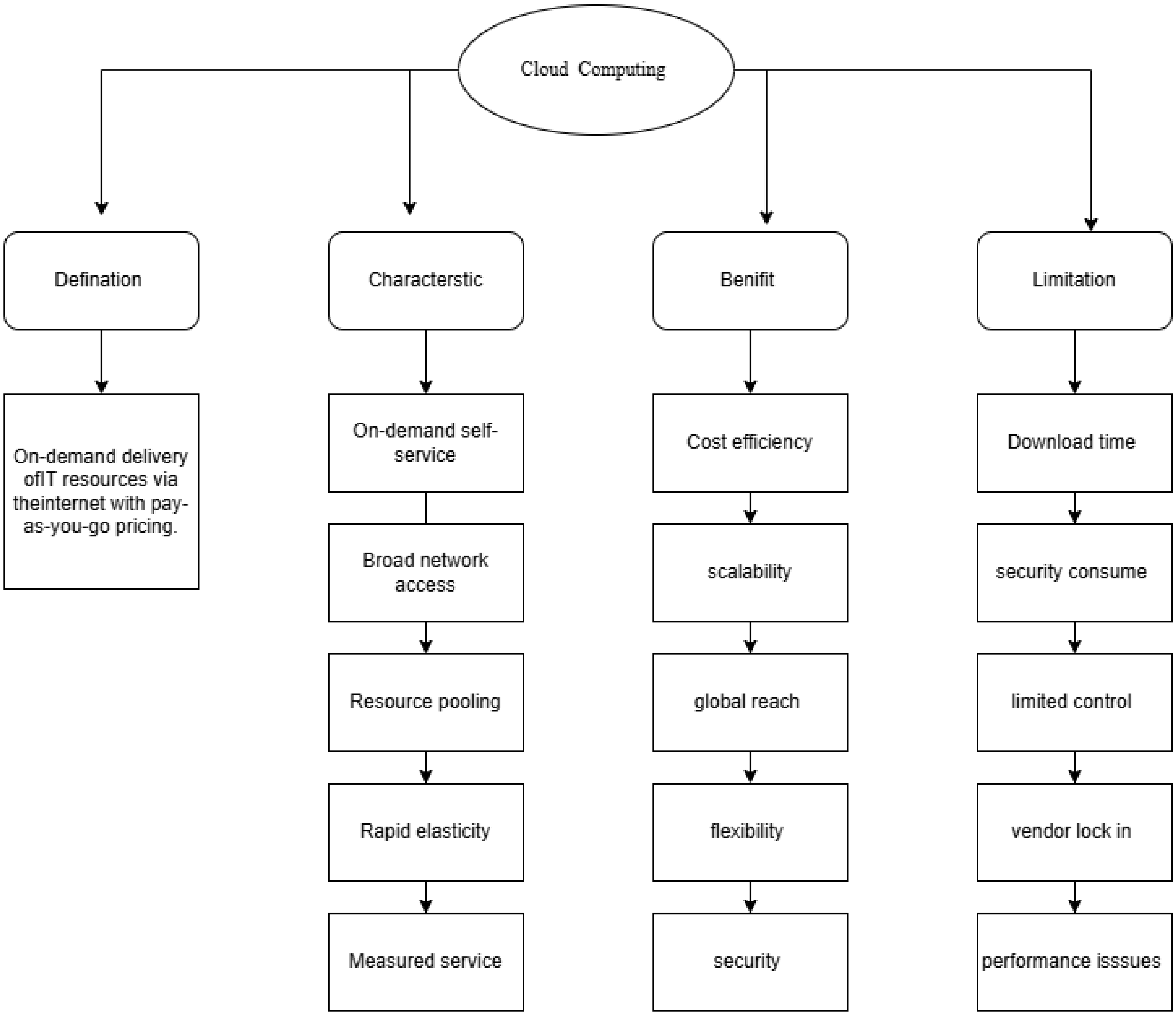
Assessment Tool Weightage: 20%

Dave's Taxonomy: Imitation (Level 1) **Total Marks: 50**
Question Count: 4

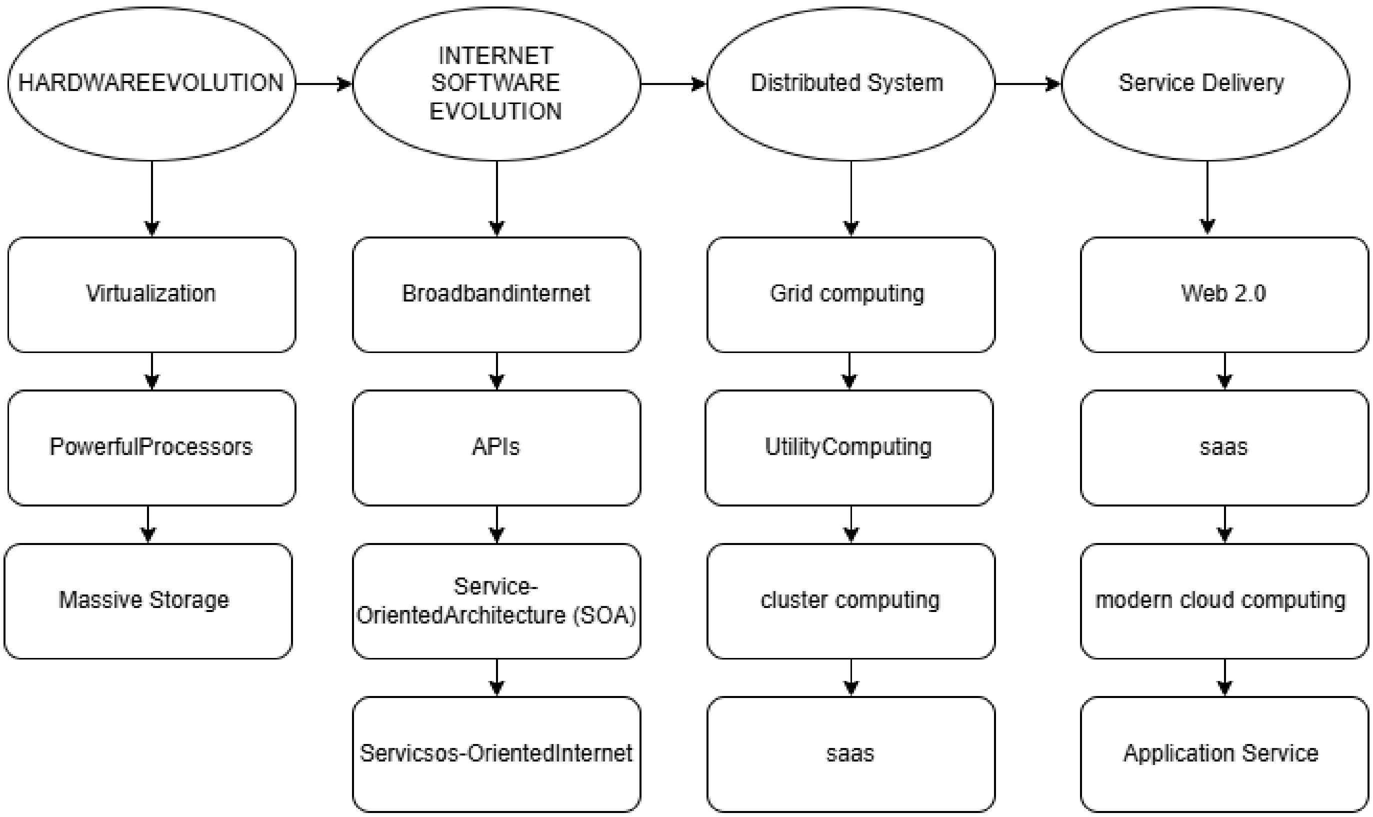
Code of Conduct
I **MAGESH.S (192525002)** certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.
I understand that any violation of academic integrity rules will result in disciplinary action.
Signature of the student: ____________________

Assessment Tasks
Concept Mapping Tasks
Create concept maps for the following tasks. Each map must show key nodes, link labels, and a logical hierarchy.

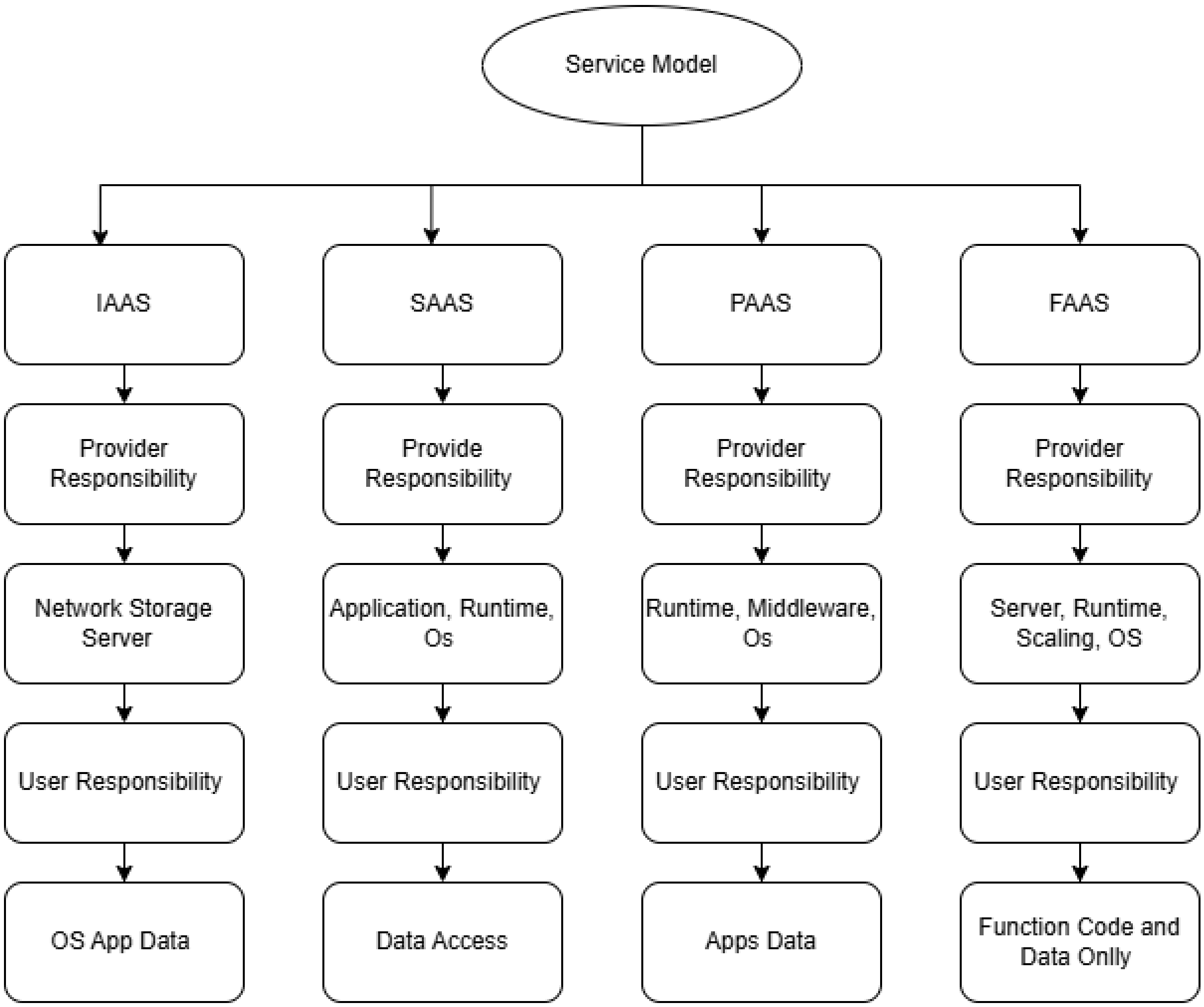
1. Concept Map 1: Cloud Computing Fundamentals. Include definition, characteristics, benefits, and limitations.



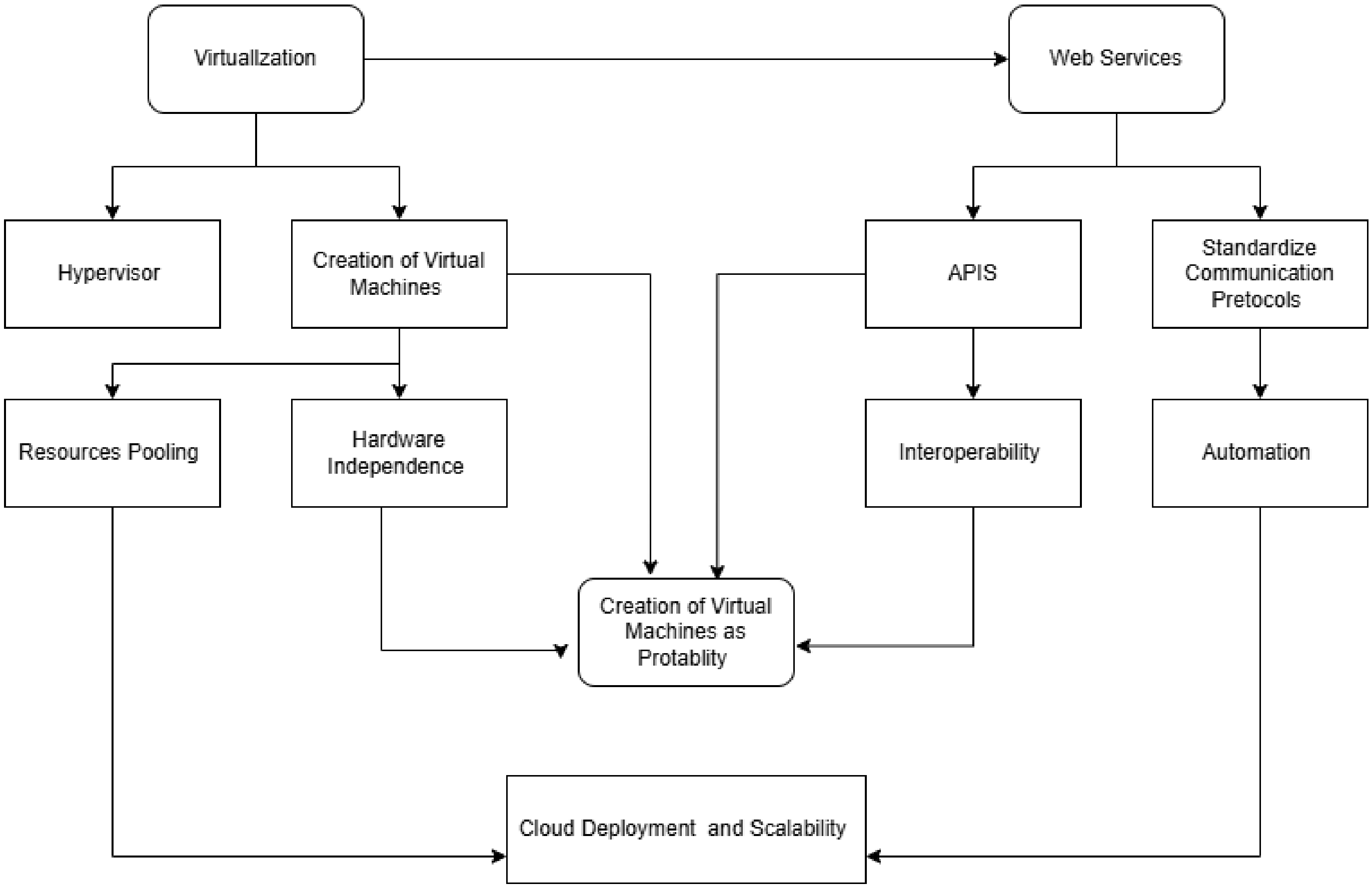
2. Concept Map 2: Evolution of Cloud Computing. Connect hardware evolution, internet software evolution, distributed systems, and service delivery.



3. Concept Map 3: Service Models. Map IaaS, PaaS, SaaS, and XaaS with provider responsibilities and user responsibilities.



4. Concept Map 4: Virtualization and Web Services as Enablers of Cloud. Show how these two concepts support cloud deployment and scalability.



Expected concept nodes:

- Elasticity, scalability, resource pooling, on-demand access, measured service
- Virtual machine, hypervisor, abstraction, API, SOAP, REST, provider, consumer
- Infrastructure layer, platform layer, application layer, shared responsibility

Concept Mapping Rubric

Criteria	Weightage	Level 4 - Exemplary	Level 3 - Proficient	Level 2 - Developing	Level 1 - Beginning
Concept Identification	25%	All major concepts are accurately identified	Most concepts are identified correctly	Limited concepts are identified	Essential concepts are missing
Relationship Mapping	25%	Links between concepts are accurate and meaningful	Most links are correct	Some links are weak or unclear	Relationships are mostly incorrect
Hierarchy and Organization	20%	Excellent structure with clear hierarchy	Mostly organized with minor issues	Basic structure with weak hierarchy	No logical organization
Technical Relevance	20%	Map strongly reflects cloud course content	Mostly aligned to topic	Partially aligned to topic	Weak alignment to topic
Visual Clarity	10%	Neat, readable, and easy to interpret	Generally readable	Somewhat cluttered	Poorly presented

5. Questions

i). Evaluate the effectiveness of your concept map in representing cloud computing principles. Which design decisions most strongly support your solution, and why? My concept map effectively represents cloud computing principles by organizing the main concepts such as on-demand access, resource pooling, scalability, elasticity, virtualization, web services, and service models in a clear hierarchy. The use of connecting links shows how virtualization and web services support cloud deployment, while IaaS, PaaS, and SaaS explain different levels of service delivery. This makes the architecture easier to understand and shows how different cloud technologies work together.
ii). Analyze how scalability, elasticity, and resource management influence the architecture shown in your concept map. What challenges could arise if these capabilities were absent? Scalability allows cloud resources to increase or decrease according to workload requirements. Elasticity enables resources to be dynamically allocated and released when demand changes. Resource pooling allows providers to share computing resources efficiently among multiple users. These capabilities improve performance and reduce unnecessary costs. If these capabilities were absent, the system could suffer from resource wastage, poor performance, service delays, overloading, and higher operational costs.
iii). Justify the selection of the APIs, web services, or cloud services included in your concept map. How do they contribute to the overall functionality and efficiency of the system? APIs and web services such as REST and SOAP are included because they enable communication between cloud applications and services. REST APIs are lightweight and suitable for modern web and cloud applications, while SOAP provides structured and standardized communication. Cloud service models such as IaaS, PaaS, and SaaS provide different levels of control and convenience. These technologies improve system functionality, interoperability, automation, and efficient access to cloud resources.

iv). Critically examine the relationships between the components in your concept map. How do these interactions improve reliability, performance, or user experience?

The relationships between the components improve the overall performance and reliability of the cloud architecture. Virtualization abstracts physical hardware and allows multiple virtual machines to share resources efficiently. Hypervisors manage virtual machines, while APIs and web services allow consumers to access cloud resources. The service models provide different levels of responsibility between providers and users. Together, these interactions support scalability, resource sharing, flexibility, reliability, and better user experience.

v). Reflect on the process of developing your concept map. How has creating and analyzing the map changed your understanding of cloud computing architecture, and what improvements would you make if

Creating the concept map improved my understanding of cloud computing architecture by showing how different concepts are connected rather than studying them separately. I understood that virtualization forms the foundation for efficient resource management, while web services and APIs enable communication between users and cloud applications. The map also helped me understand the relationship between cloud service models and shared responsibilities. If I redesigned the map, I would improve the visual hierarchy, add more real-world examples of cloud services, and use clearer connections between virtualization, web services, scalability, and service models.